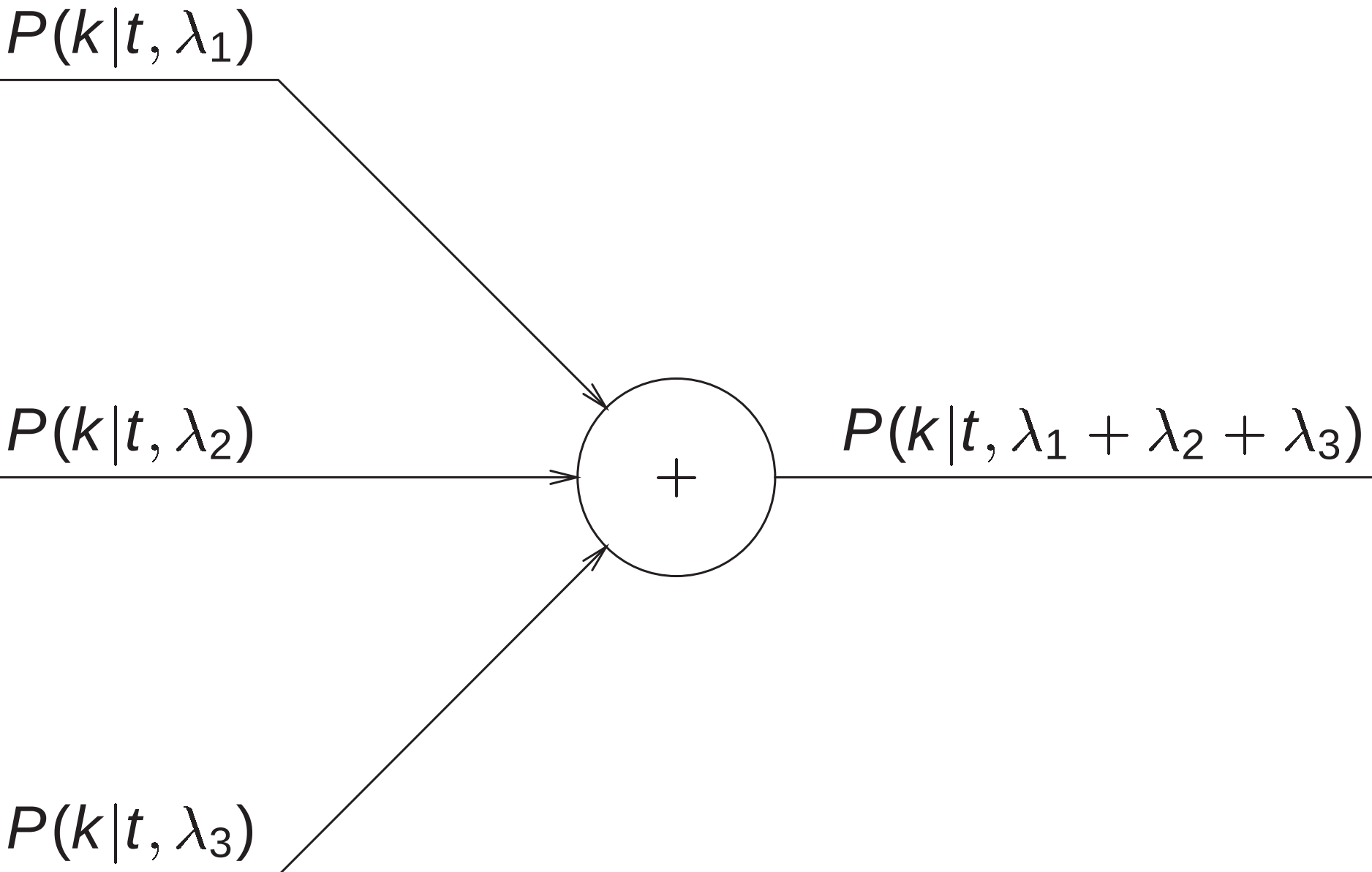
$P(k|t,\lambda_1)$
$P(k|t,\lambda_2)$
$P(k|t,\lambda_3)$
+
$P(k|t,\lambda_1+\lambda_2+\lambda_3)$